

Continuum Damage Mechanics, Anisotropy and Damage Deactivation for Brittle Materials Like Concrete and Ceramic Composites

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ABSTRACT: The main lines of a Continuum Damage Mechanics Modelling are reviewed, for applications to brittle materials treated as elastic damageable ones.

The proposed damage models consider in the same framework the cases of initially isotropic materials like concrete and anisotropic composites.

Attention is focused on the damage deactivation effects and on the possibility to describe irreversible strains directly associated to the damaging processes without any additional dissipation. Finally, the model is applied to two particular SiC/SiC and C/SiC composites.

1. INTRODUCTION

THE CONTINUUM DAMAGE Mechanics (CDM) has been greatly developed and used since the initial works of L. M. Kachanov [1]. Several domains of Continuum Mechanics and Fracture Mechanics have been significantly modified, with many new and substantial improvements. Many old and recent reviews of CDM can be mentioned, considering various classes of materials (metals, alloys, concrete, soils, ceramics, composites. . .), submitted to complex loading histories and to various kinds of physical and microstructural processes. The reader is referred for example to the papers by Kachanov [2], Hult [3], Krajcinovic [4], Lemaitre and Chaboche [5], Chaboche [6] and Lemaitre [7].

In this paper, special aspects of CDM are considered, related to the various frameworks by which the behavior of a damaged material can be postulated, still considering the damaged medium as a global continuum. The basic framework of CDM is recalled, stressing the concepts of effective stress and the ways of in-

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roducing coupling between damage and elastic behavior. Particular attention is devoted to problems related to initial or induced anisotropy and the unilateral character of the damage. Coupling between damage and plasticity is not discussed herein. The materials concerned are concrete, considered initially isotropic, and composites, in particular ceramic matrix composites. Some specific models are developed considering scalar damage variables for initially anisotropic materials.

Below, we consider only small perturbations and isothermal conditions. The total strain ϵ is assimilated to the elastic strain and the damage processes are considered time-independent. We do not discuss the difficult problems related to instabilities by localization and therefore describe only the first stages of damage, before conditions become critical.

2. BASIC CONCEPTS

The phenomenological approach of Continuum Damage Mechanics can be incorporated into a general thermodynamic framework, using internal state variable concepts [8,9]. The thermodynamic bases were recalled several times [10,11] especially in the case of a damageable elastic material. We summarize briefly the theory, using two specific potentials:

The thermodynamic potential—i.e., the Helmholtz or Gibbs free energy that supplies the state equations, including the elasticity equation. We set (T is the temperature, S is the entropy):

$$\Psi = u - TS \quad \Psi^* = \Psi - \sigma : \epsilon \quad (1)$$

$$\Psi = \Psi_*(\epsilon, d_a) \quad \text{or} \quad \Psi^* = \Psi^*_*(\sigma, d_a) \quad (2)$$

where d_a represents the family of damage variables that are scalars or tensors. Conventional processing yields:

$$\sigma = \frac{\partial \Psi}{\partial \epsilon} \quad \text{or} \quad \epsilon = -\frac{\partial \Psi^*}{\partial \sigma} \quad (3)$$

$$y_a = -\frac{\partial \Psi}{\partial d_a} \quad \text{or} \quad y_a = \frac{\partial \Psi^*}{\partial d_a} \quad (4)$$

where y_a , represents the thermodynamic force associated with the variable d_a .

The dissipation potential—corresponding here to a unique choice, since the only dissipation is due to the damage: Actually, we place ourselves in an “associated” framework in which the dissipation potential is identified as the surface limiting the undamaged domain $f \leq 0$ defined in the space of variables y_a . We set:

$$f(y_a; d_a) = g(y_a) - r(d_a) \leq 0 \quad (5)$$

$$\dot{d}_a = \dot{\mu} \frac{\partial f}{\partial y_a} \quad (6)$$

The damage multiplier is determined by the consistency condition $f = \dot{f} = 0$. Adding the loading/unloading condition yields:

$$\dot{d}_i = H(f) \frac{\left\langle \frac{\partial g}{\partial y_a} \dot{y}_a \right\rangle}{\frac{\partial r}{\partial d_\beta} \frac{\partial g}{\partial y_\beta}} \frac{\partial g}{\partial y_i} \quad (7)$$

Clearly, the heat dissipation associated with the damage growth, which reduces here to $\Sigma_a y_a \dot{d}_a$, is always positive.

3. CASE OF INITIALLY ISOTROPIC MATERIALS

3.1 Damage Definition and Effective Stress Concepts

Initially isotropic materials with no particular symmetry is the most complex case since damage is generally not isotropic. The induced anisotropy is defined on the principal axes related to the loading causing the damage (and possibly its history). The use of a second- or a fourth-rank damage tensor is then necessary. Let us first examine the two classical effective stress concepts.

The method of strain equivalence—[10] states that the effective stress tensor $\tilde{\sigma}$ expresses the stress that would have to be applied to undamaged material to cause the same strain tensor that the one observed in the damaged material submitted to the current stress σ . Using the fourth-order “*damage effect*” operator $M(d_a)$ we write:

$$\tilde{\sigma} = M^{-1}(d_a) : \sigma \quad (8)$$

In this case, the damaged material elasticity equation (completely active damage) could be written:

$$\sigma = \tilde{\Lambda}(d_a) : \epsilon \quad \epsilon = \tilde{S}(d_a) : \sigma \quad (9)$$

$$\tilde{\Lambda}(d_a) = \frac{1}{2}(M : \Lambda + \Lambda : M) \quad \tilde{S}(d_a) = \frac{1}{2}(M^{-1} : S + S : M^{-1}) \quad (10)$$

The advantage of this solution is that it does predefine the form of the damaged

elasticity operator, i.e., the relation between d and M . This method naturally leads to the use of a fourth-rank damage tensor D and to setting:

$$M(D) = I - D \quad (11)$$

where I is the fourth-rank unit tensor ($2I_{ijkl} = \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}$).

The method of energy equivalence—[12] states that the effective stress $\tilde{\sigma}$ and the effective strain $\tilde{\epsilon}$ are such that the elastic energy of the undamaged material subjected to $\tilde{\sigma}$ and $\tilde{\epsilon}$ is the same as for the damaged material subjected to σ and ϵ . We of course write $W_e = 1/2\tilde{\sigma}:\tilde{\epsilon} = 1/2\sigma:\epsilon$ and we set:

$$\tilde{\sigma} = M^{-1}(d_a):\sigma \quad \text{and} \quad \tilde{\epsilon} = M(d_a):\epsilon \quad (12)$$

It is clear that there is a certain indetermination in this case. It is resolved by the choice of the form of operator M . We can use a damage tensor d of only the second order and express $M(d)$ [13] as the following “diagonal form” in the principal coordinate system of d , using the Voigt notations:

$$M(d) = \begin{bmatrix} 1 - d_1 & & & & & \\ & 1 - d_2 & & & & \\ & & 1 - d_3 & & & \\ & & & \sqrt{(1 - d_2)(1 - d_3)} & & \\ & & & & \sqrt{(1 - d_3)(1 - d_1)} & \\ & & & & & \sqrt{(1 - d_1)(1 - d_2)} \end{bmatrix} \quad (13)$$

It should be noted right away that this operator is effectively intrinsic. It can be written alternatively like:

$$M(d) = \frac{1}{2}[(1 - d)^{1/2} \oplus (1 - d)^{1/2} + (1 - d)^{1/2} \otimes (1 - d)^{1/2}] \quad (14)$$

in which the tensor operations are defined as:

$$C = A \oplus B \rightarrow C_{ijkl} = A_{ik}B_{jl}$$

$$C' = A \otimes B \rightarrow C_{ijkl} = A_{il}B_{jk}$$

$$C'' = A \otimes B \rightarrow C''_{ijkl} = A_{ij}B_{kl}$$

With this damage effect operator, the elasticity tensors are expressed:

$$\bar{\Lambda} = M(d):\Lambda:M(d) \quad \bar{S} = M^{-1}(d):S:M^{-1}(d) \quad (15)$$

The advantage of this method is the use of a tensor of only the second order. The drawback is related to the lack of degrees of freedom. For instance, the above expression of S cannot account for the compliances of elastic media damaged by parallel cracks (solutions by Hoenig [14], by Kachanov [15,16]).

REMARKS:

1. The above definitions supply both the tensorial nature of the damage state variable, the associated measurement methodology and the effect of the damage on the constitutive behavior (elastic).
2. The “effective” stress introduced in the two above approaches should not be confused with the “net” stress defined in other theories [17] which takes into account only a reduction in resistant section, i.e., a purely geometric effect, measured by the size and density of the microcracks or microvoids. A recent paper by Murakami et al. [18] analyzes this kind of measure more in detail.
3. The particular case of isotropic damage should involve a fourth-order isotropic operator $M(d)$, not necessarily limited to the fourth-order identity tensor. We then write [19], with the strain equivalence, $M(d) = 1/3(\delta - d)1 \otimes 1 + dI$, where d is the damage variable affecting shear and δ is that affecting the hydrostatic modulus. We again have two damage variables as with Ladeveze’s [20,21]. The particular case $\delta = d$ was initially used for isotropic damage [10,5], and leads to a Poisson’s ratio that is not varying with damage [16,19].
4. Equations (10) and (11) apply generally, for material that is not necessarily initially isotropic. The symmetry of $\bar{\Lambda}$ and $\bar{S}$ is correctly preserved. It should be noted that this was not the case of the expression used until recently (Ju [19], with Equation (10), and the first author in his initial works on damage anisotropy [22]).

3.2 Choice between Helmholtz and Gibbs Free Energy

Beyond a purely thermodynamic choice with a potential depending on the elastic strains (Helmholtz) or the stresses (Gibbs), which choices are generally equivalent, the formal framework of Continuum Damage Mechanics induces significant differences between the two approaches.

This is the case in particular when the associated thermodynamic forces are included in the damage criterion (see above). With an approach based on the Gibbs potential, the damage criterion actually involves the stresses as the driving variables of the damage process. On the contrary, the Helmholtz potential naturally introduces a strain criterion. Both cases have their advantages and drawbacks:

With a potential based on stresses—involving the compliance tensor, it is relatively easy to construct a theory catering for the compliance variations obtained by simple analyses for varied microcrack configurations (micromechanical theories developed, for instance, by Kachanov [15,16] or Horii and Nemat-Nasser [23]). However, in this case the damage has to be introduced as an additive term, something like $\tilde{S} = [S:(1 + D)]$, (see, for instance Reference [19]). We then lose the concept of damage defined through a reduction in section of resistant material (effective). This approach also has the advantage of facilitating identification since the mechanical tests are generally conducted under controlled stress, in particular in the transverse directions.

A potential based on strains—has three advantages: (1) it is consistent with other analytic solutions in special cases with completely parallel cracks [19], at least when we use the strain equivalence and a fourth-rank damage tensor; (2) the direct expression of an equation between the associated thermodynamic force Y and the strain, again in the same equivalence framework. The potential is linear in D , which means the Y is independent of D . This allows significant simplifications in the structural computations for which the “local” step is always conducted with controlled strain; (3) the damage criterion based on strains allows a natural distinction between uniaxial tension and uniaxial compression. In the latter case the positive transverse strains can initiate and open cracks parallel to the compression axis. Such a compression damage criterion cannot be obtained naturally from an expression using stresses [24,25].

3.3 Unilateral Character of Damage

The damage is obviously irreversible. But its effect on the mechanical behavior can be active or inactive. In practice, depending on the applied loading, the cracks can be open or closed, leading to unilateral behavior, typically with a bilinear elasticity behavior.

This point is crucial for a damage theory with induced anisotropy, as has already been shown by Chaboche [11,26]. In effect, all the theories with anisotropic damage suggested until then had relatively serious theoretical shortcomings (see Table 1 summarizing these shortcomings for particular examples of References [24,27,28]). Either the symmetry of the elasticity operator was not respected or discontinuities of the stress-strain response under nonproportional multiaxial loadings were evidenced. “Progressive” unilateral contact functions make it possible to avoid mathematical discontinuity but in no way settle the underlying physical problem. Solution procedures proposed by Ladevèze [21] solve only partly the problem, for initially isotropic materials.

A unilateral condition respecting continuity was suggested by the first author [11,29]. It consists of writing an “opening/closing” type criterion in the current damage principal axes (axes of the equivalent microcracks) based on normal strain or normal stress (normal to these equivalent cracks, and therefore to the

Table 1. The elastic compliance matrix for biaxial principal strains with four anisotropic and unilateral damage theories. λ and μ are the Lamé's coefficients.

	$\epsilon_1 > 0$	$\epsilon_2 > 0$	$\epsilon_1 < 0$	$\epsilon_2 > 0$
Vectors [27]	$\lambda + 2\mu + 2(C_1 + C_2)\omega_1^2$ $\lambda + C_1(\omega_1^2 + \omega_2^2)$	$\lambda + C_1(\omega_1^2 + \omega_2^2)$ $\lambda + 2\mu + 2(C_1 + C_2)\omega_2^2$	$\lambda + 2\mu$ $\lambda + C_1\omega_2^2$	$\lambda + C_1\omega_2^2$ $\lambda + 2\mu + 2(C_1 + C_2)\omega_2^2$
2nd Order Tensors [28]	$h_{11}^2 + \lambda(1 - \delta)$ $h_{11}^2 + \lambda(1 - \delta)$	$h_{11}^2 + \lambda(1 - \delta)$ $h_{22}^2 + \lambda(1 - \delta)$	$h_{11}^2 + \lambda(1 - \delta)$ $h_{11}^2 + \lambda(1 - \delta)$	$h_{11}^2 + \lambda(1 - \delta)$ $h_{11}^2 + \lambda(1 - \delta)$
4th Order Tensors [24]	$(\lambda + 2\mu)(1 - d)$ $\lambda(1 - d)$	$\lambda(1 - d)$ $(\lambda + 2\mu)(1 - d)$	$\lambda + 2\lambda$ λ	λ $(\lambda + 2\mu)(1 - d)$
New Formulation				
2nd Order Tensors [25]	$(\lambda + 2\mu)(1 - d_1)^2$ $\lambda(1 - d_1)(1 - d_2)$	$\lambda(1 - d_1)(1 - d_2)$ $(\lambda + 2\mu)(1 - d_2)^2$	$\lambda + 2\mu$ $\lambda(1 - d_1)(1 - d_2)$	$\lambda(1 - d_1)(1 - d_2)$ $\lambda(1 - d_2)(1 - d_2)^2$

corresponding principal directions). The key feature of the criterion is that only the diagonal stiffness term corresponding to the normal strain, which changes sign, is modified unilaterally.

For instance, with a strain expression, we propose to write the effective elastic stiffness tensor Λ_{eff} as a function of the stiffness tensor with totally active damage $\bar{\Lambda}$ [given by equations such as (10a) or (15a)] in the following form:

$$\Lambda_{eff} = \bar{\Lambda} + \eta \sum_{i=1}^3 H(-\epsilon_i) P_i : (\Lambda - \bar{\Lambda}) : P_i \quad (16)$$

where Λ is the initial elasticity tensor, $P_i = n_i \otimes n_i \otimes n_i \otimes n_i$ is the fourth-order projection operator on the principal direction n_i , H is the Heaviside function and $\epsilon_i = n_i \cdot \epsilon \cdot n_i = \text{Tr}(P_i : \epsilon)$ is the normal strain associated with direction n_i . Coefficient η , with values between 0 and 1, plays a phenomenological role to weight the unilateral character (between the initial modulus and a compression modulus of inactive damage, different from the initial modulus). It should be noted that the double projection in the above equation indicates the diagonal term $\Lambda_{iiii} - \bar{\Lambda}_{iiii}$.

A similar formulation is available to express the effective compliance. But it should be noted that it leads to different results (S_{eff} is not equal to the inverse of Λ_{eff}). The closing criterion then naturally concerns the normal stress instead of the normal strain, with the corresponding difficulties for evidencing active compression damage with cracks parallel to the compression axis (situation illustrated in Figure 1), as already mentioned above.

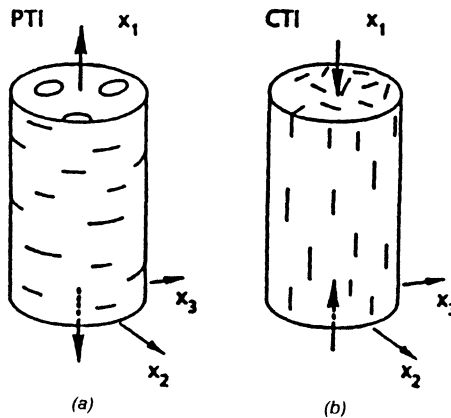


Figure 1. Two specific cracking arrangements. (a) Planar transverse isotropy, produced by $\sigma_1 > 0$, $\sigma_2 = \sigma_3 = 0$; (b) cylindrical transverse isotropy, produced by $\sigma_1 < 0$, $\sigma_2 = \sigma_3 = 0$.

Obviously, for a damageable elastic medium without residual strain (the problem of residual strain is discussed further down), we can express the Helmholtz free energy as a quadratic function of strain (by subdomains):

$$\Psi = \frac{1}{2} \epsilon : \Lambda_{eff} : \epsilon \quad (17)$$

where Λ_{eff} now depends on the current state of the damage and the sign of the normal strains ϵ_i . Recently, it was redemonstrated that this method was necessary to ensure continuity of the stress-strain response [30].

The damage criterion and the damage growth equations can be expressed as a function of the thermodynamic force Y . This force is depending on ϵ , as a continuous function (with a discontinuous derivative when the unilateral condition occurs). In a recent paper [25], we suggested a simple form of the damage criterion, for a fourth-order damage tensor with only four coefficients depending on the material. Other forms can be used, depending on the choice of the type of tensor variable, the form of the effective stress chosen, the unilateral criterion selected, etc.

4. CASE OF ANISOTROPIC MATERIAL (COMPOSITE)

We now discuss the case of composites, anisotropic materials in which the physical nature of the constituents (oriented and with very different characteristics) mean that the microcracks forming the damage are generally parallel or perpendicular to the fibers (or yarns). The development of damage thus can modify the intensity of the anisotropy (differences along the principal axes of the composite) but does not alter the initial symmetries of the material. In this case, it is reasonable to consider a simpler theory: the damage variables are scalars associated with each of the three principal directions of anisotropy of the material (assimilated here to the directions of the constituents, assumed mutually parallel or perpendicular). It should be noted that much work has already been done on the Continuum Damage Mechanics of composite materials, both in France (Ladevèze [21,31], Alix et al. [32,33], Perreux and Oytana [34], Baste et al. [35], Maire [36], and many others) and abroad. The models proposed use scalar variables in one way or another. It should be noted that the use of vectors was also suggested by Talreja [37], but it can be shown that in this particular context, this amounts to strictly the same thing. Here, we summarize a class of models proposed recently by Reference [38], particularly applied to the case of SiC/SiC and C/SiC composites.

4.1 Expression of the Thermodynamic Potential

The potential chosen is quadratic as a function of the elastic strain. Actually,

an initial strain tensor ϵ_o is introduced to describe irreversible strain (observable in C/SiC). We chose the Helmholtz free energy with the form:

$$\Psi = \frac{1}{2}(\epsilon - \epsilon_o) : \Lambda_{eff} : (\epsilon - \epsilon_o) + (\epsilon - \epsilon_o) : \Lambda_R : \epsilon_o \quad (18)$$

where Λ_{eff} is defined by the same method as for isotropic material [Equation (16) above]. This avoids any stress or strain discontinuities for any loading and any damage closing or deactivation time. In this case, directions n_i are obviously the principal directions of the material, considered invariable, and the normal strains ϵ_i are to be considered on the difference $\bar{\epsilon} = \epsilon - \epsilon_o$. Moreover Λ_R is taken as the effective stiffness with "all damage deactivated," i.e., given by (16) with $\epsilon_1 < 0$, $\epsilon_2 < 0$, $\epsilon_3 < 0$. We obtain the stress by differentiation:

$$\sigma = \Lambda_{eff} : (\epsilon - \epsilon_o) + \Lambda_R : \epsilon_o \quad (19)$$

As is shown schematically in Figure 2(b), the behavior obtained by this theory shows the degradation of the elastic modulus in tension, with a unilateral closing effect for a particular strain state ϵ_o , to which corresponds the stress state:

$$\sigma_o = \Lambda_R : \epsilon_o \quad (20)$$

The modulus on the compression side ($\epsilon_1 < \epsilon_{o1}$) is not necessarily equal to the initial modulus Λ . This depends on the other strains, transverse, and the Poisson's ratios of the damaged material. It should also be noted that the components of tensor ϵ_o are not necessarily the same [39].

With this choice of potential, the existence of a residual strain ϵ_R is described without introducing any additional dissipation. Similar approaches were suggested by Hermann and Kestin [40], Ramtani [28], Saanouni et al. [39,41], but

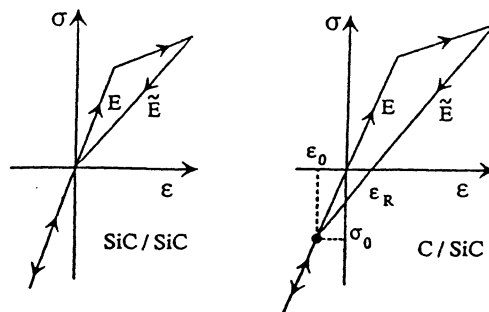


Figure 2. Schematic behavior of SiC/SiC and C/SiC.

without evidencing the exact closing point defined by $\epsilon = \epsilon_0$. The residual strain is obtained easily as:

$$\epsilon_R = [I - \Lambda_{eff}^{-1} : \Lambda_R] : \epsilon_0 \quad (21)$$

It is of course directly related to the damage involved in Λ_{eff} .

In the framework in which we placed ourselves, the damage variables are assumed scalar, d_1, d_2, d_3 , corresponding to the crack densities in the three principal axes of the material. We use the energy equivalence, with Equation (15a) to define the elasticity operator $\bar{\Lambda}$, using a damage effect tensor M supplied by Equation (11):

$$\bar{\Lambda} = (I - D) : \Lambda : (I - D) \quad (22)$$

in which the damage effect tensor D , a fourth-rank tensor, does not have a "diagonal" form. In the principal system of the material, we set:

$$D = \begin{bmatrix} d_1 & \alpha_{12}(d_1 + d_2) & \alpha_{13}(d_1 + d_3) & & & \\ \alpha_{12}(d_1 + d_2) & d_2 & \alpha_{13}(d_2 + d_3) & & & \\ \alpha_{13}(d_1 + d_3) & \alpha_{13}(d_2 + d_3) & d_3 & & & \\ & & & \alpha_{44}(d_2 + d_3) & & \\ & & & & \alpha_{44}(d_1 + d_3) & \\ & & & & & \alpha_{66}(d_1 + d_2) \end{bmatrix} \quad (23)$$

REMARKS:

1. The closing criterion obviously involves the normal values (normal to the principal directions) of the strains $\epsilon = \epsilon_0$. We deliberately chose a strain criterion (instead of a stress criterion) to be able to describe the initiation of microcracks parallel to the axis of a simple compression loading.
2. The form chosen for the damage effect tensor is more general than in the case of isotropic material (with energy equivalence and a second-order damage variable). The scalar variables simplify the expression, making the theory more flexible.

4.2 Damage Criterion

The three thermodynamic forces can be associated with the three scalar variables d_a . These forces are easily expressed from Equations (18), (16) as:

$$y_a = -\frac{1}{2} \bar{\epsilon} : \frac{\partial \bar{\Lambda}}{\partial d_a} : \bar{\epsilon} + \eta \sum_i H(-\bar{\epsilon}_i) \bar{\epsilon}_i^2 \left(P_i :: \frac{\partial \bar{\Lambda}}{\partial d_a} \right) - \bar{\epsilon} : \frac{\partial \Lambda_R}{\partial d_a} : \epsilon_0 \quad (24)$$

in which the sign $::$ denotes the doubly contracted product of two fourth-order tensors (the result has a scalar value). In addition, (22) allows us to write:

$$\frac{\partial \bar{\Lambda}}{\partial d_a} = -[\Gamma_a::\Lambda:(I - D) + (I - D):\Lambda:\Gamma_a] \quad (25)$$

where $\Gamma_a = \partial D / \partial d_a$ is easily obtained by differentiating (23).

The damage criterion is expressed in the standard way through a surface defined in the space of variables y_a . Two types of choices are possible:

1. Multiple criteria, a priori decoupled:

$$f_a = \langle y_a \rangle - r_a(d_a) \leq 0 \quad (26)$$

in which the threshold functions r_a are decoupled. They are scalar functions of d_a determined by the variation of d_a as a function of y_a and therefore implicitly as a function of the strains. It should be noted that in woven composites, directions 1 and 2 are identical (warp and woof directions), meaning that $r_1 = r_2$

2. A combined criterion in which we define a norm of vector $\{y\}$ and a norm of vector $\{d\}$:

$$\|y\| = (\langle y_1 \rangle^m + \langle y_2 \rangle^m + a \langle y_3 \rangle^m)^{1/m} \quad (27)$$

$$\|d\| = (d_1^2 + d_2^2 + d_3^2)^{1/2}$$

$$f = \|y\| - r(\|d\|) \leq 0 \quad (28)$$

As is shown in Figure 3, in a biaxial representation, exponent m adjusts the amount of coupling and coefficient a sets the relative magnitude of direction 3 (the much lower strength in this direction leads to taking $a \gg 1$).

REMARKS:

1. It is always possible to combine these two kinds of criteria. An extension of the multiple decoupled criteria consists of taking multiple combined criteria with functions $r_a(d_1, d_2, d_3)$.
2. The introduction of undamaged domains $f \leq 0$ is replaced in certain theories [33] by the use of variables such that $y(\tau) = \sup_{\tau \leq t} y(\tau)$. It is easy to show that this, theoretically amounts to the same thing.
3. In the two above criteria, there is no increase in the damage variables associated with negative values of y_a as can also be seen from Figure 3. It also induces that damage rate can never be negative.

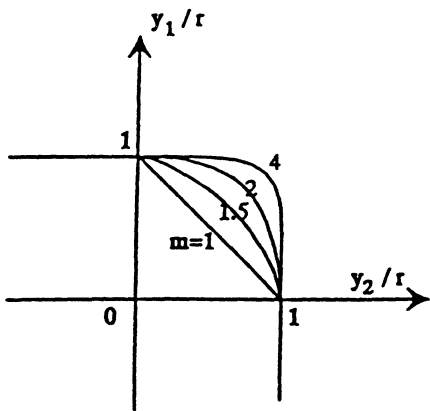


Figure 3. Schematics of the damage criterion.

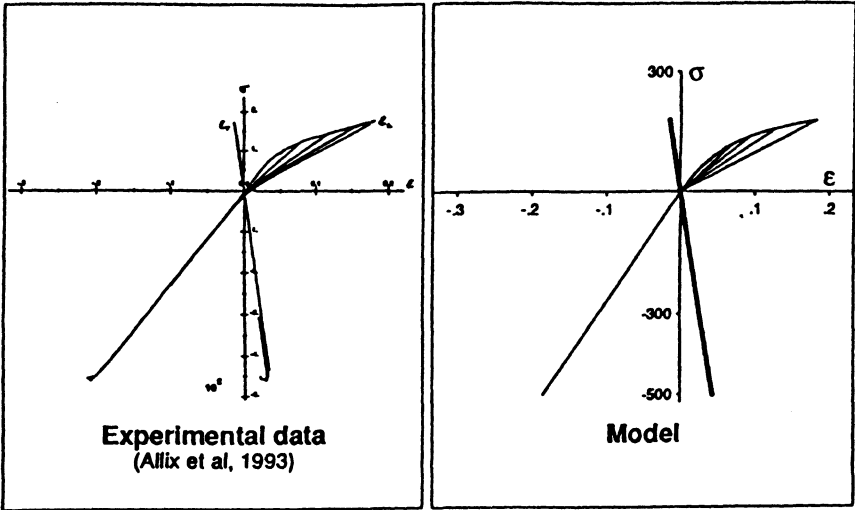


Figure 4(a). Tensile-compression test on SiC/SiC.

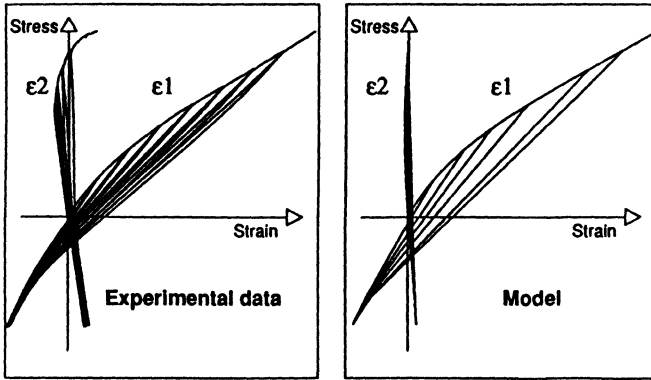


Figure 4(b) Tensile-compression test on C/SiC.

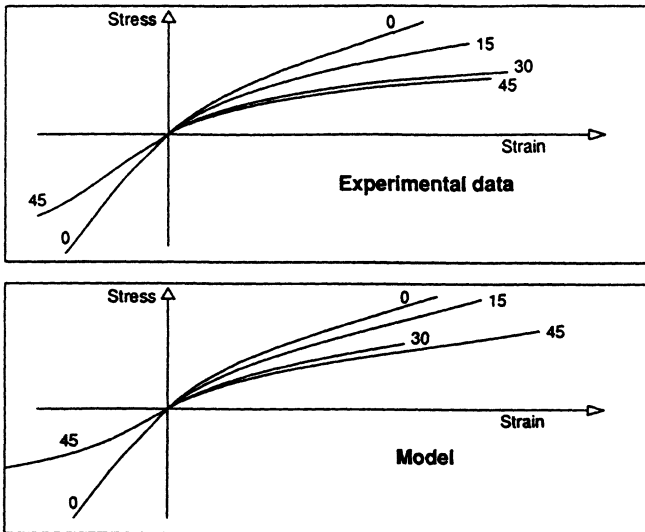


Figure 5. "Off-axis" tests on C/SiC.

4. With the uniaxial compression (along 1, for instance), damage is however possible, since we have $\epsilon_1 < 0$ but $\epsilon_2 > 0$ and $\epsilon_3 > 0$, which generates $y_2 > 0$ and $y_3 > 0$. The corresponding damage is relative to the directions crosswise to the compression direction.

4.3 Examples of Application

The theoretical model consisting of state Equations (18) and (23) and the second criterion (28) above was applied to ceramic matrix composites SiC/SiC and C/SiC (with $m = 2$). In the case of SiC/SiC, the residual strain was considered negligible (with $\epsilon_0 = 0$).

Figure 4 shows the results of uniaxial tension in direction 0 for SiC/SiC and C/SiC. The tension curves with unloading are correctly reproduced by the model, including the residual strain on C/SiC, and the closing effect is clearly visible (these curves were used for identification). The transverse strain (ϵ_2) is also correctly predicted in both cases. The compliance S_{12} is constant for SiC/SiC and variable for C/SiC (in this case, the Poisson's ratio varies with damage).

Figure 5 shows the predictions made for tension and compression tests in directions off the axes for C/SiC and their comparison with experimental data. The predictions are fairly good. Moreover, all the nine elasticity constants of the damaged material (under tension) are also correctly reproduced [42] by comparison with ultrasonic measurements made by Baste et al. [43,44] (results not shown here).

5. CONCLUSION

As was seen above, the models proposed account very correctly for the nonlinear behavior observed on composites with a ceramic matrix. They describe elastic behavior coupled with the damage as well as the associated residual strains and the damage deactivation effects, unilateral effects, that are clearly observed.

It is interesting to note that this model for composites is a particular case of a more general model of initially isotropic damageable elastic materials such as concrete. The simplification is due to the scalar character of the state variables describing the damage, related to the fact that the microcracks are considered to be guided by the heterogeneous, oriented structure of the material.

Future work will be focused on modeling the following aspects:

1. Validation of the possibilities of modeling compression damage.
2. Development of a theory accounting for damage oriented both by the initial anisotropy of the material and by the current applied loading.
3. Verification of the models and damage criteria by multiaxial experiments with nonproportional loading.

4. Taking into account the irreversibilities not directly associated with the damage, with additional dissipation.
5. Taking into account the viscosity and hysteresis effects observed in certain composites.
6. Proposal and validation of damage growth laws for initially isotropic materials.

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